www.tntextbooks.in

2.9.3. Determinants of Elasticity of Demand

There are many factors that determine the degree of price elasticity of demand. Some of them are described below:

a) Availability of Substitutes:

If close substitutes are available for a product, then the demand for that product tends to be very elastic. If the price of that product increases, buyers will buy its substitutes; hence fall in its demand will be very large. Hence, price elasticity will be larger. Eg. Vegetables.

For salt no close substitutes are available. Hence even if price of salt increases the fall in demand may be zero or less. Hence salt is price inelastic.

b) Proportion of consumer's income spent' if smaller proportion of consumer's income is spent on particular commodity say X, price elasticity of demand for X will be smaller. Take for example salt, people spend very small proportion of their income on salt. Hence, salt will have small elasticity of demand, or inelastic.

c) Number of uses of commodity:

If a commodity is used for greater number of uses, its price elasticity will also be larger. For example, milk is used as butter milk, curd, ghee and for making ice cream etc. Hence, even the small fall in the price of milk, will tempt the consumers to use more milk for many purposes. Hence milk has greater price elasticity of demand.

d) Complementarity between goods:

For example, along with petrol, lubricating oil is also used for running automobiles. Here, a rise in the price of lubricating oil may not reduce the demand for lubricating oil. Hence, the complementary good, here, lubricating oil, will be price inelastic.

e) Time: In the long run, the price elasticity of demand for many goods will be larger. This is so because, in the long run many substitutes can be discovered or invented. Therefore, the demand is generally more elastic in the long run, than in the short run. In the short run bringing out new substitutes is difficult.

2.9.4 Measurement of Elasticity of Demand

There are three methods of measuring price elasticity of demand.

1. The Percentage Method

$$e_p = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$

It is also known as ratio method, when we measure the ratio as:

$$e_p = \frac{\%\Delta Q}{\%\Delta P} \text{ where,}$$

$\%\Delta Q$= percentage change in demand

$\%\Delta P$ = Percentage change in price

2. Total Outlay Method

Marshall suggested that the simplest way to decide whether demand is elastic or inelastic is to examine the change in total outlay of the consumer or total revenue of the firm.

Total Revenue = (Price x Quantity Sold)

TR = (P x Q)